ENGINEERING GROWTH

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Abstract
This paper offers the first systematic historical evidence on the role of a central actor in modern growth theory: the engineer. We construct a database on the share of engineers in the labor force during the Second Industrial Revolution (1870–1914) at the county level for the United States and the state and national levels for the Americas. These measures are robustly correlated with income today after controlling for literacy, other types of higher-order human capital (college graduates, lawyers, physicians, patenting) and demand-side factors, as well as after instrumenting engineering using the 1862 US Land Grant Colleges program. Differences in engineering density in 1880 accounted for 10% of the higher US county incomes today, while national disparities in engineering density can explain approximately a quarter of the income divergence in the Americas. To document the mechanisms through which engineering density works, we show how it is correlated with higher rates of technology adoption and structural transformation across intermediate time periods and with numerous measures of the knowledge economy today. (JEL: O11, O30, N10, I23)

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Teaching Slides
A set of Teaching Slides to accompany this article are available online as Supplementary Data.

"You have all the elements, but you cannot make steel."
—Andrew Carnegie (1900)[1]

1. Introduction

Carnegie's taunt to the owners of the Birmingham Steel Company suggests the difficulties the US South faced in absorbing the knowledge required, including inventing the necessary modifications to local context, to establish dynamic industries and converge to the North (Wright 1986). This remains the case in today's developing regions: a vast literature now puts barriers to technological adoption at the center of explanations of the distribution of income globally.[2] An absence of human capital is often seen as impeding technology adoption and invention, yet there is no consensus on what type of human capital is most important and, in particular, the role of the "upper tails" of the education distribution, including higher-order technical skills. Endogenous growth theory (Romer 1990), empirical studies of technological transfer and innovation, as well as historical accounts put the engineer and associated scientific institutions at the center of the growth process, especially during the Second Industrial Revolution (1870–1914) (Mokyr 1998). Yet, to date, there is no systematic evidence of the historical prevalence of engineers and, hence, their plausible importance to explaining the patterns of development that we see.

In the present paper, we first make this basic data contribution. We establish the stylized facts surrounding the relative density (labor share) of engineers during the Second Industrial Revolution at the county level for the United States, at the sub-national level for six countries in the Americas and at the national level for the Western Hemisphere and relevant benchmark countries of Europe. The engineering data collection is done in a systematic way, drawing on graduation records, membership in professional societies, and census data.

We characterize the late 19th century engineer as possessing the formal university level training necessary to design and build public infrastructure and the machinery and systems needed by the private sector. The ability to read scientific journals and blueprints, and in general scope the global technological frontier and adapt it to the local context, makes the engineer a primary vector of knowledge transmission as well as contributor of new ideas. Hence, we also see our estimates as capturing indigenous *technical capacity* (or what Nelson (1993) terms *innovative capability*) defined as the

1. Cited in Wright (1986, p. 171).

2. See Parente and Prescott (1994), Eaton and Kortum (1999), Eaton and Kortum (2001), Caselli and Coleman (2001), Keller (2004), Klenow and Rodriguez-Clare (2005), Comin, Hobijn, and Rovito (2008), Comin and Mestieri (2018), and Section 1.1.

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ability to absorb and invent new technologies during the second wave of the Industrial Revolution, a period when science became more formalized and technological adoption required higher level and more specific skills.[3]

To empirically establish the importance of such capacity, we employ our rich US county-level data set to establish a robust relationship between engineering density and modern income, controlling for an extensive set of geographical, growth related, and human capital variables that may be confounded with engineering. We further instrument for the possible endogeneity of engineering using the 1862 US Morrill Land Grant Colleges program. This program had an important long-run impact in establishing engineering training institutions; however, this was not its initial intent, and the location of these colleges was largely independent of demand considerations—as we show empirically.

We also incorporate geo-located patenting density collected by Acemoglu et al. (2018) as a measure of technical capacity more related to invention. Akcigit, Grigsby, and Nicholas (2017) identify a causal relationship between patented inventions and long-run economic growth, which raises the question of whether patents are capturing the same effect as engineers. We find this *not* to be the case: including patents diminishes the impact of engineers by only 15%. While engineering density has some impact through patenting, the two measures are capturing different things: patents may be capturing a particular human capital—perhaps natural scientists—more dedicated to *invention* (Griliches 1990; Khan 2005); engineering density captures more *adoptive* technical capacity (see Mokyr 1992). We then demonstrate plausible mechanisms through which engineering density worked by looking at technology adoption and structural change at intervals across the following century, as well as the incidence of high-tech industries and within-industry technical upgrading today.

Both innovation variables show an important and robust effect on income: differences in engineering density in 1880 account for a 10% rise in county-level income today, and patenting capacity contributes another 10%. Further, we find very similar coefficients at the "state" level both for the US—suggesting migration issues among geographical units are not critical here—and the sub-national panel.

We then document the extraordinary variance in engineering density across countries of very similar income levels in 1900—for instance, Argentina, Chile, Denmark, Sweden, and the US South—which correlates well with their divergent income positions today (see Figure 1). Further, the similarity of Latin American values with those of Spain and Portugal, and those of the United States with Northern Europe, suggests an important colonial inheritance of technical capabilities.[4] We then use historical evidence to validate our ranking of technical capacity and, through case studies, document how—consistent with the dynamics of the Second Industrial

3. While economic historians have long debated about whether the First Industrial Revolution was a deskilling process (see, most recently De Pleijt, Nuvolari, and Weisdorf 2020, Hanlon 2022), the Second Industrial Revolution was decidedly skilled. See Gordon (2016) for a discussion of the centrality of this period to United States and world growth.

4. We explore the determinants of the innovation measures in Online Appendix Tables G.16 and G.17.

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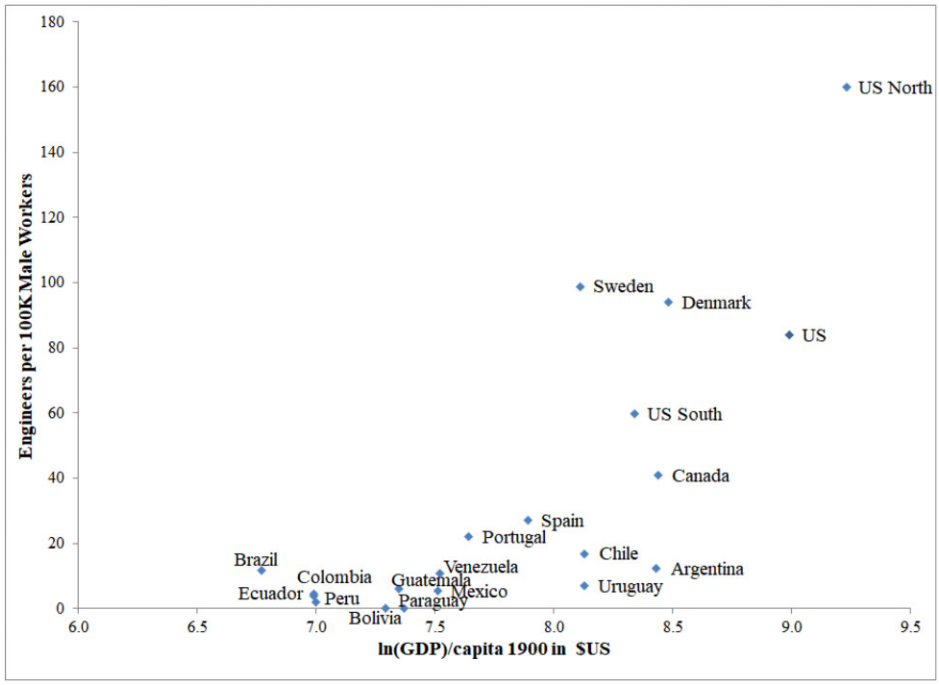


FIGURE 1. Income in 1900 and engineering density in 1900. Plot of engineering density versus Log of GDP per capita in 1900 from Maddison. Engineering density is accumulated graduates of engineering programs per 100,000 male workers around 1900, as described in Online Appendix A.

Revolution—technological progress and positive development outcomes require the more professional engineering and scientific capabilities that we quantify. Applying the well-estimated US county-level coefficient to the national-level data, the national differences in engineering density account for up to a quarter of income differences today.

1.1. Literature

The economics literature on technology diffusion and human capital has a long pedigree. Nelson and Phelps (1966) were the first to model this relationship formally, Sala-i-Martin and Barro (1995) related it to growth convergence patterns, and Benhabib and Spiegel (2005) generalized their theoretical framework. Parente and Prescott (1994), Eaton and Kortum (1999), Eaton and Kortum (2001), Keller (2004), and Klenow and Rodriguez-Clare (2005) expanded on the impact of technology adoption and development, both theoretically and empirically. Most recently, Comin, Hobijn, and Rovito (2008) and Comin and Mestieri (2018) argue that the diverging use of new technologies plausibly explains observed TFP differentials and can drive simulations that closely track the magnitudes of the Great Divergence of the last two centuries (e.g. Pritchett 1997; Galor and Weil 2000; Galor 2011). Empirically, Foster and Rosenzweig

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(1995, 1996, 2010) focus on the adoption of agricultural technologies in India, while Caselli and Coleman (2001) study the adoption of computers. Both are channels to growth that we will explore.[5]

The literature has postulated several *types* of human capital that may be important for growth and technology adoption. Literacy and accumulated years of schooling or enrollment have received the most attention (Krueger and Lindahl 2001; Glaeser et al. 2004; Barro and Lee 2015, although as Vandenbussche, Aghion, and Meghir (2006), Aghion et al. (2009), and Gennaioli et al. (2013) note, the composition among levels of education matters as well, depending on distance from the frontier.[6] In his classic article on endogenous growth, Romer (1990) highlights the "research engineer" and Mokyr (2005), the minority of "trained engineers, capable mechanics and dexterous craftsmen on whose shoulders the inventors could stand" (see also Meisenzahl and Mokyr (2011, p. 16)). Rosenberg (2000) and Nelson (2005) stress the accumulated ability and scientific institutions to manage new ideas for innovation and invention, and Cohen and Levinthal (1989) and Griffith et al. (2004), the capacity for research and development needed for technological transfer.

Several authors, including Mokyr (1998) and Howitt and Mayer-Foulkes (2005), stress that higher-order human capital and the institutions that generated and housed it may have had an even more determinant role at the dawn of the Second Industrial Revolution, which saw an increased emphasis on more structured scientific inquiry, such as laboratory-based R&D. This scientifically oriented human capital and a technologically savvy entrepreneurial class were necessary to tap into the expanding and increasingly sophisticated global stock of knowledge and convert it into local growth. The technological leap forward also meant an erosion in the efficacy of existing levels of human capital and technical capacity, relative to that needed to continue to adopt (see Howitt 2000; Aghion, Howitt, and Mayer-Foulkes 2005). Building on this Schumpeterian insight, Howitt and Mayer-Foulkes (2005) argue for multiple equilibria in innovation where countries whose human capital evolved with the frontier at the time of the technological leap forward could innovate or adopt, but those whose frontier-adjusted human capital did not keep up slipped into an equilibrium where even the adoption of technologies was difficult, and stagnation followed.

Among *empirical* attempts to document the impact of even very basic measures of human capital, Hanushek et al. (2017), in a spirit similar to the present work, find that differences in human capital account for 20%–35% of the current variation in GDP per capita among US states (see also Hsieh et al. 2019). However, there have been relatively few efforts to systematically capture what kind of human capital matters

5. A recent development literature has emphasized that the main role of human capital in economic growth is to facilitate the adoption of technology from abroad rather than serve as an independent factor of production. See Mayer (2001) and Asif and Lahiri (2019).

6. Other dimensions figure importantly as well: Lucas (1993) and Young (1993), among others, stress the importance of accumulated "learning by doing"; Zeev, Mokyr, and Van Der Beek (2017) and De la Croix, Doepke, and Mokyr (2018) focus on apprenticeship; and Baumol (1990) on entrepreneurial skills and orientation.

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or even to document the stocks of different types of capital. At the most basic level Valencia Caicedo (2018) shows how Jesuit missionaries improved literacy, positively impacting modern incomes. Goldin and Katz (2011), Goldin (1999), and Goldin and Katz (1999) have famously traced the evolution of secondary and tertiary education in the context of US growth. Judson (1998) and Castelló-Climent and Mukhopadhyay (2013) attempt to assess whether tertiary education matters more or less than primary schooling. Perhaps closest to this work, Murphy, Shleifer, and Vishny (1991) document that countries with a higher proportion of engineers today grow faster relative to those with more law concentrators. They present suggestive evidence that for the present period, there exists a high correlation between income and the density of engineers, though at this stage, causality remains elusive.

Historically, Squicciarini and Voigtländer (2016) show that mid-18th century French cities with more subscriptions to the *Encyclopédie*—proxying for higher-order human capital—grew faster after the dawn of the First Industrial Revolution. Becker, Hornung, and Woessmann (2011) and Cinnirella and Streb (2017) show that human capital was also crucial for later Prussian innovation. Universities, as generators of higher-order human capital, have also been shown to contribute to growth (Cantoni and Yuchtman 2014; Valero and Van Reenen 2019). In China, Yuchtman (2017) shows that during the early 20th century, engineers enjoyed massive wage premia over all other classes of higher education, and that modern Western education and engineers in particular were critical for China's ability to adopt Western technologies. Waldinger (2016) shows that the dismissal of scientists in Nazi Germany led to a permanent decrease in growth, while Moser, Voena, and Waldinger (2014) and Nunn, Sequeira, and Qian (2020) show that foreign immigration to the United States substantially increased invention.

To date, however, neither the historical prevalence of Romer's research engineers nor Mokyr's "engineers and mechanics" have been quantified in a globally comparable form, in particular during periods of intense technological change—such as the Second Industrial Revolution—in the United States and the Americas at large. We seek to remedy this gap with new data as well as by offering estimates of the importance of engineers to the growth process, providing a new proxy for *adoptive* technical capacity which is conceptually closer to the upper tails of scientific knowledge.

2. Data

The paper focuses on the divergence within the Western Hemisphere, and hence the data effort concentrates primarily on the United States, Canada, and Latin America. At the aggregate level, however, we also collect national data on Portugal and Spain to see how far the colonial apple may have fallen from the imperial tree. Additionally, we collect data on two countries that have been paired with Latin American countries as studies in divergence: Denmark and Sweden, which, in 1900, shared similar levels of size, income per capita, and status as natural resource exporters (Figure 1). Online Appendix Figure F.1 includes Germany and France as well, for reference. Table 1

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TABLE 1. Summary statistics (county level, United States).

Variable	Mean	p50	Standard deviation	Min	Max	Observations
Ln income	10.044	10.032	0.219	9.165	11.360	1905
Engineers	22.22	0.000	37.56	0.000	299.19	1905
Patents	0.450	0.273	0.588	0.000	5.336	1905
Ln LGC distance	0.175	0.281	0.682	−5.450	1.765	1905
Rainfall	0.077	0.076	0.042	0.009	0.393	1905
Altitude	0.025	0.023	0.019	0.000	0.175	1905
Ruggedness	0.005	0.003	0.006	0.000	0.043	1905
Distance to river	0.017	0.014	0.014	0.000	0.169	1905
Average temperature	−0.002	−0.006	0.061	−0.160	0.188	1905
Distance from coast	0.027	0.023	0.020	0.000	0.102	1905
Population density	0.003	0.001	0.049	0.000	2.055	1905
Manufacturing GDP	0.042	0.018	0.066	0.000	0.576	1904
Slavery	0.153	0.019	0.215	0.000	0.925	1905
Farms per capita	0.105	0.110	0.038	0.000	0.698	1905
Railroads	0.584	1.000	0.493	0.000	1.000	1905
Literacy	0.760	0.829	0.203	0.151	1.000	1905
School assistance 12–17	0.051	0.055	0.018	0.000	0.090	1905
Educational score	0.073	0.073	0.019	0.035	0.168	1905
Lawyers	107.35	100.18	55.14	0.000	490.92	1905
Physicians	153.58	150.39	61.99	0.000	420.35	1905

Notes: Log income per capita in 2000. Engineering density measured by engineers per 100,000 male workers. Patents density measured by patents per 100 habitants. Distance to LGC is distance to Land Grant College measured by distance between the near LGC and the county centroid. Population density is the number of individuals per 100 square kilometers in 1880. Literacy share of the population that is literate in 1880. Secondary is share of 14–17-year-olds who report that they are attending school. Tertiary is the percentage of workers who report completing one or more years of college. Farms per capita is the number of farms per capita retrieved from the University of Virginia Library. Railroads is an identifier variable if the county has railroad. Slavery to county level is taken from Nunn (2008). Lawyer density is measured by lawyers per 100,000 individuals. Physicians' density is measured by physicians per 100,000 individuals. Mining is total mining output in 1860 was hundred thousand dollars. Rivers captures the distance between the near river and the county. Rivers taken from HydroSHEDS (USGS 2011). Temperature is a yearly average in degrees celsius, altitude measures the elevation of the capital city of the state in kilometers, and rainfall captures total yearly rainfall in meters. Distance to coast is the distance between the coast and the county centroid; ruggedness of Terrain from Nunn and Puga (2012). Manufacturing output is the manufacturing output per capita in 1880, taken from NHGIS.

presents the summary statistics at the US county level and Table 2 at the state and country levels for the Americas.

2.1. Measuring Engineering

We collect what we believe to be the first comparable historical engineering data at three levels of aggregation: county for the United States, sub-national (state), and national. In each case, our measure of technical human capital is the number of engineers per 100,000 male workers (since there are very few female engineers at the time). Greater detail and the raw data are available in the Online Appendix. Table 3 summarizes the different sources, and Online Appendix A discusses how sources of data were combined for each country in detail.

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TABLE 2. Summary statistics (state level, Americas).

Variable	Mean	p50	Standard deviation	Min	Max	Observations
Ln income	9.03	8.92	0.91	7.13	11.18	377
Engineers	23.38	11.00	30.08	0.00	84.00	275
Engineers (sub)	82.07	39.68	105.70	0.00	472.59	170
Population density (1900)	41.06	4.44	243.20	0.00	3319.27	235
Literacy	40.85	34.00	23.95	11.30	86.70	337
Literacy (sub)	49.06	39.96	30.10	4.60	98.31	175
Railroads	3.15	1.80	2.71	0.30	9.30	377
Railroads (sub)	65.12	48.54	57.46	5.16	309.20	49
South	0.15	0.00	0.36	0.00	1.00	111
Slavery	20.67	3.28	25.16	0.00	72.66	83
Lawyers	218.92	139.22	210.93	1.64	1156.44	114
Mining output	0.47	0.12	1.06	0.00	6.49	45
Spain	0.81	1.00	0.39	0.00	1.00	390
Land suitability	0.56	0.58	0.28	0.00	1.00	384
River density	3.28	3.29	1.23	0.00	6.92	386
Average temperature	19.97	20.40	5.83	2.38	29.00	332
Rainfall	1.28	1.10	0.95	0.00	8.13	332
Altitude	0.66	0.19	0.92	0.00	4.33	332
Distance from coast	0.87	0.91	0.12	0.45	1.00	383
Ruggedness	126.89	99.33	103.53	0.00	474.34	378

Notes: Log income per capita in 2000 (Purchasing Power Parity (PPP) 2005 US dollars). Engineering density is measured by engineers per 100,000 male workers. Population density is the number of individuals per square kilometer in 1900. Pre-colonial population density measures the number of natives per square kilometer in 1492. Literacy share of the population that is literate in 1900. Literacy share of the population that is literate in 1900, sub-national. Railroad density is measured as miles of track per 1,000 square kilometers. Railroad density is measured as miles of track per 1,000 square kilometers, sub-national. South is a dummy variable for whether the US state is a southern state according to the US Census. Slavery is measured as a fraction of the population and is taken from Bergad (2008) and Nunn (2008). Lawyer density is measured by lawyers per 100,000 individuals. Mining is the total mining output in 1860 was hundred thousand dollars for the US states. Spain is a dummy for whether the country was a Spanish colony: Argentina, Chile, Mexico, and Venezuela. Agriculture suitability is an index of the probability of cultivation given cultivable land, climate, and soil composition, from Ramankutty, Foley, and McSweeney (2002). Rivers capture the density of rivers as a share of land area derived from HydroSHEDS (USGS 2011). Temperature is a yearly average in degrees celsius, altitude measures the elevation of the capital city of the state in kilometers, and rainfall captures total yearly rainfall in meters, all are from Bruhn and Gallego (2011). Distance from the coast from Gennaioli et al. (2013); ruggedness of Terrain from Nunn and Puga (2012).

County Level. For the US county-level data, we use the census of 1880 to generate the density of engineers, lawyers, and medical doctors.[7] During this early time period, the data on engineering sub-specialties are sparse. Still, we present a correlation table between engineering specialties in Online Appendix Table G.2. This type of information is not available for other countries during this historical time period.

7. As an interesting historical note, the 1880 US census data were collected under Francis Walker as superintendent, the first President of the American Economic Association.

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TABLE 3. Sources of the engineering data.

Country	Data source	National density	Geographic level	Source
Argentina	Graduation records	12	National	Centro Argentino de Ingenieros (1981)
	1895 census data		Sub-national	1895 census
Bolivia	Book	0	National	Contreras (1993)
Brazil	Graduation records	12	National	Telles (1994)
	Periodical records		National	Almanaque Laemmert
	Engineering association records		National	Almanak dos Engenheiros
Canada	Graduation records	41	National	McInnis (2004)
Chile	Graduation records	17	National	Serrano (1993) and Villalobos et al. (1990)
	1907 census data		Sub-national	1907 census
Colombia	Graduation records	5	National	Facultad de Ingeniería (2011)
	Engineering association records		National	Anales de Ingeniería
	Book		National	Safford (1976)
	1912 census data		Sub-national	1912 census
Denmark	Graduation records	92	National	Harnow (1997)
Ecuador	Graduation records	2	National	Escuela Politécnica Nacional
France	Book	176	National	Ahlström (1982)
Germany	Book	170	National	Ahlström (1982)
Guatemala	Graduation records	6	National	Universidad de San Carlos de Guatemala
Mexico	Graduation records	5	National	Bazant de Saldaña (1993)
	Engineering association records		National	Association de Ingenieros
	1895 census data		Sub-national	1895 census
Paraguay	Graduation records	0	National	Universidad Nacional de Asunción
Peru	Graduation records	5	National	Lopez Soria (2012)
Portugal	Engineering association records	22	National	Associação dos Engenheiros Civis Portuguezes
Spain	Engineering schools records	12	National	Riera i Tuebols (1993)
	Engineering and technical schools records	42	National	López et al. (2005)
Sweden	Graduation records	90	National	Ahlström (1993)
United States	Graduation records	84	National	Mann (1918) and Edelstein (2009)
	1880 census data		National	1880 census
	1900 census data		Sub-national	1900 census
Uruguay	Graduation records	7	National	Universidad de la República, Uruguay
Venezuela	Engineering association records	11	National	Mendez (2013)
	1926 census data		Sub-national	1926 census

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Sub-National Level. Argentina, Chile, Colombia, Mexico, Venezuela, and the United States. We collect census data at the sub-national (state) level for six countries in the hemisphere, allowing us to create a spatial panel. A drawback is that it is the individual respondent who decides what profession to declare and there is limited institutional validation. To address the potential heterogeneity in the de facto definition, we include state and country fixed effects. Data are derived from the Argentine Census of 1895, Mexican National Census of 1895, Chilean Census of 1907, Colombian Census of 1912, Venezuelan Census of 1926, and US Census of 1900.[8] The other Latin American countries did not have data available for this time period at this degree of disaggregation.

National Level. Argentina, Bolivia, Brazil, Canada, Chile, Colombia, Denmark, Ecuador, Guatemala, Mexico, Paraguay, Portugal, Spain, Sweden, Uruguay, Venezuela, and the United States. We aspire to generate one credible data point for each of the Western Hemisphere countries and the four reference countries (Table 3). To reduce the bias introduced by self-declaration of profession to the degree possible, calculations are done with actual graduates of domestic engineering colleges and universities within the country. Though most countries also employ foreign engineers, we are interested in broader indigenous technical capacity, local university engineering training, and research programs, so we focus on domestically trained engineers.[9] From the graduation flows, we then generate the stocks at the turn of the century. Since the working life of an engineer is roughly 40 years, we begin accumulating the stock in 1860, discounting it in each period by 0.983 as the rate of annual death/attrition. In some cases, we have a long series of graduation records that make this procedure straightforward. In Bolivia, Denmark, Mexico, New York, Peru, Spain, Sweden, Venezuela, and Mexico, the flow of graduated engineers is available for the 1860–1900 period. As in most of the Latin American cases, the division of engineer into sub-categories (civil, mechanical, etc.) occurred only in the 20th century, and our sources generally do not disaggregate by type, hence we work with the broad umbrella category of "engineer". For rough orders of magnitude, we refer to estimates for the frontier countries of France and Germany by Ahlström (1982), which were generated similarly, although since we did not construct them, we did not use them in our analysis. In Argentina, Brazil, Chile, Colombia, Ecuador, and the United States as a whole, graduation series are less complete and we bring other sources of data to bear.

Data on membership in engineering societies or official registries validate the broad orders of magnitude of our generated stocks where there is overlap. In Brazil, for instance, registry with the government was required to be a practicing engineer, and this is an important input in our measure. In the other countries, membership in these was not required, so registration and hence likely underestimates the true value. However,

8. Online Appendix Figure F.2 shows the high degree of correlation between the 1880 county and 1900 state engineering variables for the United States.

9. For Argentina and Chile, we have data on both foreign and domestic engineers and including the former leaves the results unchanged. We also control for the number of foreigners in Online Appendix Table G.17 and find no significant impact on the results.

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together with the fragmentary graduation data, they help narrow down a credible stock measure. For many countries, there were simply no engineering programs or even universities by 1900, and we did not pursue them. As Online Appendix A documents, in Central America and the Caribbean, only Guatemala offers evidence of an established engineering school with documented graduates. To get an aggregate measure for the United States, we calibrate the census data with early national graduate estimate records as well as detailed graduate records for New York. In our exploratory cross-country empirics, we include only those countries for which we have the necessary covariates, which, in general, are not available for countries below 1 million inhabitants in 1900. Table 3 summarizes the national numbers and sources.

2.2. Patenting

For the United States, we also construct a measure of county level average patenting activity from 1890–1910, drawing on patents granted by the United States Patent and Trademark Office (USPTO) and provided by Acemoglu et al. (2018). This is likely to capture technical capacity more related to invention, which began to assume a central role in the Second Industrial Revolution. Comparable patent data are not available for the Latin American countries, and hence the variable only enters into the county-level US analysis.

2.3. Sub-National Income per Capita

Modern county-level US income in dollars is mean household income taken from the 2000 US Census. Income in 2005 Purchasing Power Parity (PPP) US dollars is drawn from a highly disaggregated spatial data set on population, income, and poverty, built on the basis of national census data by the World Bank (2009) for the World Development Report on *Reshaping Economic Geography*.[10]

2.4. Control Variables

As control variables we use: literacy, secondary schooling, higher-level non-engineering human capital, railroads, mining, manufacturing output, population density in 1880, slavery, a US South dummy, agricultural intensity, and a set of geographic and weather controls including temperature, altitude, rainfall, agricultural suitability, river density, distance to coast, and ruggedness. Online Appendix B offers details on the construction and description of these measures.

2.5. Mechanisms

As for intermediate historical outcomes, we look at various measures of technological progress and structural change, among them: value added in manufacturing,

10. For further details, see Maloney and Valencia Caicedo (2016).

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horsepower per manufacturing establishment, and the number of retail and financial establishments in 1930, as well as numerous present-day high-tech industry measures from the County Business Patterns (2012), Cermeño (2019), and Lexis/Nexis Ci Technology Database (2016).

3. Empirical Results

3.1. US County-Level Data

To establish the link between historical technical capacity and present-day income, we employ our engineering and patenting density data for the United States at the county level.[11] Table 4 presents the Ordinary Least Squares (OLS) estimates including the groups of controls (in bold, below) for geography, economic activity, and other human capital measures. All specifications have robust errors and are clustered at the state level, and are estimated either with OLS or fixed effects. In particular, we estimate

$$Y_{2000,i} = \alpha + \gamma_E Eng_{1880,i} + \gamma_P Pat_{1880,i} + \boldsymbol{Geo_i \gamma_{Ge}} + \boldsymbol{Y_{1880,i} \gamma_{Gr}} + \boldsymbol{HC_{1880,i} \gamma_H} + \mu_{state} + \varepsilon_i. \quad (1)$$

To be consistent with the other samples, we begin with an OLS specification of engineering density along with a vector of geographical controls. We report beta coefficients to ease interpretation and comparison across covariates. In Table 4, column (1), which uses our maximum sample, engineering emerges strongly, significantly, and of the expected positive sign in the OLS specification. Temperature and distance from the coast both enter negatively and significantly. Column (2) replicates the specification but with the smaller sample dictated by the more restrictive coverage of slavery and education in subsequent exercises, and shows that any changes in the results with their addition are not due to sample changes. Going forward, changes in the reported sample size are due to the availability of the corresponding covariates.

Table 4, column (3) then includes the set of income-related controls that together work to reduce the correlation that may occur simply because historically denser, richer areas with more infrastructure, and in particular, manufacturing and railroads, as in Hornbeck and Rotemberg (2019), may be correlated with engineers in 1880 and with income today through non-innovation related channels.[12] This helps us distinguish what engineers could have built, in terms of infrastructure, from their innovation-related activities. We further include a measure of slavery as an institutional variable that may be correlated with low human capital, but also provides other channels to present

11. A few counties change their borders in the last century. This and missing observations leave us with 1,905 observations. Regressions with the maximum number of observations using the harmonization procedure in Hornbeck (2010) are presented in Online Appendix Table G.1, which leaves our results unchanged. Our baseline results also hold without controls.

12. For instance, the presence of railroads is positively correlated with engineers, not shown.

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TABLE 4. Engineering capacity as a determinant of income (county level, United States).

Dependent variable: Ln (income per capita in 2000)								
	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS	(6) OLS	(7) OLS	(8) OLS
Engineers	0.224***	0.288***	0.190***	0.148***	0.184***	0.126***	0.117***	0.0930***
	(0.0338)	(0.0403)	(0.0350)	(0.0287)	(0.0329)	(0.0284)	(0.0270)	(0.0247)
Rainfall	0.0263	0.0332	0.0298	0.0553	0.0290	0.0720	0.0816	0.132*
	(0.0643)	(0.0741)	(0.0695)	(0.0707)	(0.0700)	(0.0640)	(0.0713)	(0.0774)
Altitude	−0.00325	−0.0183	0.0347	0.00555	0.0425	0.0186	−0.00955	−0.0714
	(0.0334)	(0.0758)	(0.0730)	(0.0784)	(0.0737)	(0.0714)	(0.0747)	(0.0654)
Ruggedness	−0.0817	−0.0825	−0.104	−0.0298	−0.115	−0.0770	−0.0335	−0.0201
	(0.0828)	(0.106)	(0.0911)	(0.0850)	(0.0894)	(0.0783)	(0.0788)	(0.0766)
River distance	0.0236	0.0142	0.0210	0.0181	0.0183	0.0329	0.0282	0.0293
	(0.0324)	(0.0311)	(0.0296)	(0.0271)	(0.0285)	(0.0286)	(0.0268)	(0.0276)
Temperature	−0.281***	−0.311***	−0.122*	−0.0101	−0.0787	−0.120*	−0.0314	−0.0820
	(0.0488)	(0.0501)	(0.0624)	(0.0534)	(0.0661)	(0.0664)	(0.0658)	(0.162)
Coast distance	−0.171***	−0.237***	−0.171***	−0.130**	−0.175***	−0.185***	−0.135***	−0.0734
	(0.0547)	(0.0489)	(0.0473)	(0.0527)	(0.0479)	(0.0467)	(0.0483)	(0.0896)
Population density			0.0930***	0.0952***	0.0934***	0.0830***	0.0895***	0.0994***
			(0.00725)	(0.00671)	(0.00735)	(0.00882)	(0.00780)	(0.00839)
Manufacturing output			0.197***	0.176***	0.213***	0.185***	0.171***	0.120***
			(0.0443)	(0.0396)	(0.0430)	(0.0446)	(0.0416)	(0.0390)
Slavery			−0.1000	0.180	−0.0418	−0.0149	0.163	0.177**
			(0.0757)	(0.109)	(0.0770)	(0.0841)	(0.109)	(0.0692)
Railroads			0.162***	0.113***	0.144***	0.131***	0.110***	0.0948***
			(0.0343)	(0.0321)	(0.0327)	(0.0313)	(0.0306)	(0.0200)
Farms per capita			0.0100	0.00698	0.00230	0.0401	0.0259	−0.00313
			(0.0616)	(0.0442)	(0.0589)	(0.0596)	(0.0476)	(0.0341)
Literacy				0.462***			0.393***	0.502***
				(0.103)			(0.104)	(0.0721)
Secondary					0.119**		0.00151	0.0374
					(0.0517)		(0.0530)	(0.0518)
Tertiary						0.143***	0.0481	0.0571
						(0.0524)	(0.0434)	(0.0397)
Lawyers						0.136***	0.115***	0.120***
						(0.0389)	(0.0371)	(0.0346)
Physicians						−0.00169	−0.0464	0.0182
						(0.0438)	(0.0396)	(0.0300)
Observations	2,380	1,904	1,904	1,904	1,904	1,904	1,904	1,904
	0.190	0.241	0.318	0.366	0.323	0.351	0.377	0.490
State FE	No	No	No	No	No	No	No	Yes

Notes: Dependent variable is log sub-national income per capita (2000). Engineering density is measured by engineers per 100,000 male workers. Patents density is measured by patents per 100 inhabitants. Rainfall captures total yearly rainfall in meters, altitude measures the elevation of the capital city of the state in kilometers, ruggedness of Terrain from Nunn and Puga (2012), rivers capture the distance between the near river and the county. Rivers taken from HydroSHEDS (USGS 2011). Temperature is a yearly average in degrees celsius; distance to coast is the distance between the coast and the county centroid. Population density is the number of individuals per 100 square kilometers in 1880. Manufacturing output is the manufacturing output per capita in 1880, taken from NHGIS. Slavery to county level is taken from Nunn (2008). Farms per capita is the number of farms per capita retrieved from the University of Virginia Library. Railroads is an identifier variable if the county has a railroad. Literacy is the share of the population that is literate in 1880. Secondary is the share of 14–17-year-olds who report that they are attending school. Tertiary is the percentage of workers who report completing one or more years of college. Lawyer and physician density is measured per 100 individuals. Beta coefficients. Robust and clustered SE in parenthesis. State level fixed effects. Column (1) employs our maximum available sample. Column (2) repeats the same regression with the sample dictated by the restricted availability of slavery and education to show that subsequent changes are not due to sample reduction. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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income. Population density, a measure of lagged economic activity (manufacturing output), and railroads enter positively and significantly and will remain so throughout the analysis. Slavery, which enters significantly and negatively with just geographical controls (not shown), loses significance. As a measure of agricultural activity, farms per capita never enter significantly. We employ alternative farming and population density controls in Online Appendix Table G.3 with no change in the results.

The next 3 columns seek to ensure that the engineering measure is not simply a proxy for human capital more generally.[13] Column (4) includes literacy, which enters positively and significantly, as expected, and reduces the engineering coefficient by just over 10%. Column (5) then includes a measure of secondary education, whose importance across the next century Goldin (1999) and Goldin and Katz (2011) have stressed, and which enters significantly and positively. Goldin and Katz (1999) also discuss the importance of the expansion in higher education across the earlier 1890–1940 period, and in column (6), we include several variables to ensure that engineers are not picking up higher-order human capital more generally: in this case, the share of individuals who report attending at least one year of college, the density of lawyers, and the density of physicians. The first two enter significantly and positively. That physicians do *not* confirms that there are types of human capital whose density is associated with income but whose effect on it is more tenuous,[14] while the significant relationship we find between engineering and income is conceptually more direct.

Finally, column (7) includes all the human capital variables together. Literacy and lawyers remain significant, although this is no longer the case with secondary education or the college variable. Either literacy or college attendance can eliminate secondary education, suggesting that, at this point in history, the latter is broadly proxying for literacy, or college readiness. The same remains true in column (8), which includes state fixed effects, which effectively control more local unobservables. We test formally for the role of such unobservable factors using Altonji, Elder, and Taber (2005) ratios, finding that the role of unobservables would have to be larger than the one of observables to explain away our result. We further explore this possibility later when we employ Least Absolute Shrinkage and Selection Operation (LASSO) techniques to select our control set, which suggests that at this point we might be over controlling in the full OLS specification.

In all cases, engineering retains its significance at the 1% level, although losing some of its magnitude from the initial specification as we progressively control for unobserved correlates. The relative robustness of engineering versus secondary education suggests that a small, well-educated technological class was, at this time, more important than broad-based post-literacy skills. The consolidation of higher education in engineering occurred well before the major expansion of secondary education, which (Goldin 1999; Goldin and Katz 2011) document rose from only

13. Online Appendix Figure F.3 plots the engineering variable against the other human capital variables: literacy, secondary schooling, tertiary schooling, and patents.

14. For physicians, this time is also before the 1910 Flexner report, which standardized the discipline.

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18% in 1910 to 71% in 1940. Across our period, the corresponding average for the country is 7%, with a maximum of 26%. Literacy—at the lower tail—is critical to growth, but beyond it, a high level of technical capacity is also necessary.[15]

The positive effect of lawyers may arise, as posited by Cantoni and Yuchtman (2014), because in earlier periods they were factors complementary to scientific human capital—critical to the establishment of institutions and economic rules of the game. If we divide the sample into those above and below the median level of lawyers (Online Appendix Table G.4), it appears that a higher density of lawyers may enhance the effect of engineers. The negative effect of lawyers found by Murphy, Shleifer, and Vishny (1991) may arise because, as institutions firmed up over time, the social value of lawyers fell as they became more oriented towards unproductive activities. The now positive and significant impact of slavery suggests that its earlier negative effect could have worked through the human capital channel, though we cannot make any causal claims at this point.

Columns (1) and (2) of Table 5 repeat the last two full OLS specifications but replace engineering density with patenting density to capture the effect documented by Akcigit, Grigsby, and Nicholas (2017) and Hanlon (2022) of inventive capital and growth. Patenting enters significantly at the 1% level again in all specifications. The coefficient is also remarkably close to that found by Akcigit, Grigsby, and Nicholas (2017) using an alternative approach. Columns (3) and (4) combine patenting with engineering density. Both coefficients decline 15%–20% in magnitude, suggesting that one channel through which engineering density operates is through patenting and that some of the apparent effect of patenting is perhaps in fact working through non-patenting channels.[16] A more formal Sobel–Goodman decomposition test shows that only 15% of the effect of engineers on income works via patents. However, that each continues to enter very significantly and independently suggests that they are capturing *different* types of technical capacities and different underlying human capital: Engineers may be more involved with technological adoption, while natural scientists or specialized infrastructure may drive the more inventive activity captured by patenting (see again Mokyr 1992). We fully accept the argument of Moser (2013) that much innovation, including patentable invention, is done outside the patent system, and hence such distinctions cannot be drawn too sharply. Though uninstrumented, taken together, the engineering and patenting coefficients can explain around 20% of the county's income differences.

15. On this point, see Galor and Moav (2006) and Franck and Galor (2022). Heterogeneous results reveal a higher impact of engineers on above the median literacy counties, not reported, evincing the importance of human capital complementarities.

16. Griliches (1998) notes the limitations of patenting as capturing inventive activity when he says that "Not all inventions are patentable, not all inventions are patented, and the inventions that are patented differ greatly in quality, in the magnitude of inventive output associated with them." Empirically, Online Appendix Table G.5 confirms that engineering density is significantly correlated with patenting, providing a mechanism through which the engineering variable is acting. Still, in the complete specification, including all measures of formal human capital, we only explain 50% of the variance.

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TABLE 5. Engineering capacity as determinants of income: patents & IV (county level, United States).

Dependent variable: Ln (income per capita in 2000)						
	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) IV	(6) IV
Engineers			0.0989***	0.0789***	0.112***	0.108***
			(0.0257)	(0.0227)	(0.0255)	(0.0241)
Patents	0.137***	0.127***	0.116***	0.113***	0.131***	
	(0.0383)	(0.0363)	(0.0377)	(0.0353)	(0.0360)	
Rainfall	0.0864	0.148*	0.0804	0.149*	0.153*	0.132*
	(0.0755)	(0.0840)	(0.0743)	(0.0814)	(0.0829)	(0.0777)
Altitude	−0.0394	−0.0994	−0.0371	−0.0951	−0.108**	−0.0806
	(0.0736)	(0.0600)	(0.0749)	(0.0628)	(0.0547)	(0.0559)
Ruggedness	−0.000979	0.00270	−0.00915	−0.000511	−0.00560	−0.0278
	(0.0755)	(0.0687)	(0.0773)	(0.0725)	(0.0669)	(0.0707)
River distance	0.0295	0.0267	0.0254	0.0247	0.0315	0.0372
	(0.0267)	(0.0262)	(0.0254)	(0.0258)	(0.0272)	(0.0293)
Temperature	−0.0306	−0.0837	−0.0368	−0.0807	−0.0690	−0.0715
	(0.0654)	(0.156)	(0.0624)	(0.155)	(0.151)	(0.159)
Coast distance	−0.120**	−0.0512	−0.124**	−0.0498	−0.0372	−0.0658
	(0.0501)	(0.0846)	(0.0496)	(0.0845)	(0.0846)	(0.0895)
Population density	0.0888***	0.0985***	0.0889***	0.0984***	0.0949***	0.0961***
	(0.00739)	(0.00793)	(0.00774)	(0.00826)	(0.00834)	(0.00838)
Manufacturing output	0.128**	0.0720	0.123**	0.0695*	0.0632	0.124***
	(0.0489)	(0.0442)	(0.0471)	(0.0411)	(0.0452)	(0.0438)
Slavery	0.176	0.187**	0.177	0.193***	0.172**	0.153**
	(0.114)	(0.0717)	(0.111)	(0.0699)	(0.0693)	(0.0693)
Railroads	0.117***	0.103***	0.119***	0.103***	0.0880***	0.0792***
	(0.0304)	(0.0202)	(0.0302)	(0.0198)	(0.0189)	(0.0193)
Farms per capita	0.000461	−0.0223	0.0275	−0.00203	−0.0315	−0.0368
	(0.0476)	(0.0345)	(0.0481)	(0.0340)	(0.0318)	(0.0325)
Literacy	0.409***	0.516***	0.399***	0.518***	0.514***	0.495***
	(0.107)	(0.0725)	(0.107)	(0.0731)	(0.0718)	(0.0704)
Secondary	0.0142	0.0488	0.0115	0.0472	0.0341	0.0233
	(0.0540)	(0.0521)	(0.0531)	(0.0518)	(0.0539)	(0.0540)
Tertiary	0.0324	0.0437	0.0251	0.0383	0.0399	0.0633*
	(0.0416)	(0.0379)	(0.0425)	(0.0383)	(0.0360)	(0.0375)
Lawyers	0.112***	0.113***	0.0937**	0.0980***	0.106***	0.135***
	(0.0399)	(0.0370)	(0.0385)	(0.0354)	(0.0389)	(0.0377)
Physicians	−0.0493	0.0166	−0.0440	0.0162	0.0146	0.0170
	(0.0390)	(0.0299)	(0.0384)	(0.0298)	(0.0293)	(0.0291)
Observations	1,904	1,904	1,904	1,904	1,904	1,904
R-squared	0.377	0.492	0.383	0.496		
State FE	No	Yes	No	Yes	Yes	Yes
F-statistical (robust)					13.75	13.75
F Cragg–Donald					19.95	19.95

Notes: Dependent variable is log sub-national income per capita (2000). Standardized beta coefficients. Engineering density is measured by engineers per 100,000 male workers. Patents density is measured by patents per 100 inhabitants. Rainfall captures total yearly rainfall in meters, altitude measures the elevation of the capital city of the state in kilometers, ruggedness of Terrain from Nunn and Puga (2012), and rivers captures the distance between the near river and the county. Rivers taken from HydroSHEDS (USGS 2011). Temperature is a yearly average in degrees celsius; distance to coast is the distance between the coast and the county centroid. Population density is the number of individuals per 100 square kilometers in 1880. Manufacturing output is the manufacturing output per capita in 1880, taken from NHGIS. Slavery to county level is taken from Nunn (2008). Farms per capita is the number of farms per capita retrieved from the University of Virginia Library. Railroads is an identifier variable if the county has a railroad. Literacy share of the population that is literate in 1880. Secondary is the share of 14–17-year-olds who report that they are attending school. Tertiary is the percentage of workers who report completing one or more years of college. Lawyer and physician density is measured per 100 individuals. Engineers instrumented using log distance to nearest Land Grant College measured by the distance between the near LGC and the county centroid, estimated by 2SLS. Beta coefficients. Robust and clustered SE in parenthesis. In column (5), the *F* statistic of the instrument's relevance is 13.75 with robust standard errors and 7.62 with robust and clustered to state level standard errors. State level fixed effects. Cragg–Donald *F*-statistic is 19.99. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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3.2. Instrumental Variables Results

Our engineering variable is already lagged a century, and in addition, we have included numerous controls for other plausible channels through which it may be correlated with present income beyond the capacity to manage and generate new technologies. We also show later that there is no correlation between engineers in 1880 and manufacturing in 1860, suggesting that earlier economic activity was not driving engineering density at this time. Nonetheless, for robustness, we also instrument our engineering measures with the log distance to the nearest Morrill Land Grant College. Moretti (2004) employs a similar instrument for general education and Kantor and Whalley (2019) show that the agricultural research stations affiliated with Land Grant Colleges raised agricultural productivity (although the impact dissipated after 20 years). Hence, though we believe that the instrument is more appropriate to the current context, we cannot definitively rule out other channels, and thus we treat the exercise as informative about robustness rather than conclusive.

The exogeneity of the location of the universities has been discussed extensively (Moretti 2004). The Morrill Program was introduced in 1862 precisely to remedy the perceived shortfalls in technical assistance in agricultural and mechanical innovation in the abstract. Its motivation was driven by both democratic ideals of educating workers during the post Civil War era and by the observation that the United States had none of the agricultural and technical schools found in Europe (Nevins 1962; Nienkamp 2010). However, location was not locally demand driven: as Johnson (1981) argues, "On the contrary, a case could be made that the new colleges were created by reformers, not practitioners, and for an ideal, not for an established need," and under-subscription remained the rule for decades.

Though the promoters of the Morrill Program did not envisage it as fomenting engineering per se, it would finance the first engineering departments in the West, Midwest, and the South, and would become a central driver in developing the higher-order scientific and engineering capacity of the country, as we detail in the Historical Section. Nienkamp (2010) in *Land-Grant Colleges and American Engineers* argues that the schools "provided the foundation, both in training and number, for twentieth-century American professional engineering". Similarly, Nevins (1962) argues that the Morrill Program "promoted the emergence of the most effective engineering schools on the globe". Seely (2005) confirms that by 1900, the majority of engineers were graduating from land grant colleges: 76% of mechanical engineers, 63% of civil engineers, and 63% of electrical engineers.[17] There is therefore a tight *a priori* link from the land grant colleges' instrument to engineering density.

17. Goldin and Katz (1999) note that while the majority of new universities of the time were privately started and Morrill cannot explain the expansion of higher education, it was central to engineering. As of 1908, they note that 60% of the nation's engineering students were found in public universities and the "geographical dispersion of engineering students came mainly from those enrolled in public schools," the rise of which they attribute to the Morrill Program (Goldin 1999).

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TABLE 6. Correlates of distance to Land Grand College: first stage (county level, United States).

	Engineers	*Lawyers*	*Physicians*
Distance LGC 1862	−0.102***	−0.0580	0.0130
	(0.0371)	(0.0497)	(0.0416)
Observations	1,904	1,904	1,904
R-squared	0.010	0.003	0.000
Distance LGC 1890	−0.0456	0.0139	0.0356
	(0.0348)	(0.0499)	(0.0402)
Observations	1,904	1,904	1,904
R-squared	0.002	0.000	0.001

Notes: Engineering density is measured by engineers per 100,000 male workers. Lawyer and physician density is measured per 100 individuals. *Distance LGC 1862 (1890)* is the log of distance between the near Land Grand College in 1862 (1890) and the county centroid. Robust and clustered SE in parenthesis. *** $p < 0.01$.

Consistent with the historical narrative, the first stage of a basic 2SLS (Table 6) reveals a strong positive correlation between engineering density and the Morrill variable with very high Cragg–Donald *F*-tests (1% level) and acceptable *F*-tests, suggesting a strong instrument. However, to exclude other possible channels through which it might be working, we undertake several additional exercises. Table 6, first row, shows that there is no significant correlation with the other measures of higher education: Lawyers and physicians, suggesting that the instrument is not capturing general higher education as in Moretti (2004), but rather technical human capital. Additionally, there is little evidence of the instrument being correlated with pre-existing trends.[18] Hence, Table 6 also shows that none of the human capital measures for 1880 are correlated with the second wave of land grant universities established in 1890, as might be expected if the location of the universities were driven by economic factors that engendered higher-order human capital independently.[19]

Columns (5) and (6) of Table 5 present the second stage results for the complete specification. Engineers enter again at the 1% level with a modest increase in value to 0.11 in the fixed effect specification, both with and without patents. The IV estimates are only slightly larger than the OLS, perhaps reflecting a small reduction in bias due to measurement error in our measure of technical human capital. It is possible, for example, that "engineer" in the 1880 census included people who did not have advanced standardized technical education, and hence our instrument will mitigate such misreporting. Still, we view the stability between the OLS and IV coefficients positively. Overall, it appears that the saturation of the OLS specification with numerous controls for economic activity may have also reduced any residual

18. Online Appendix Tables G.6 and G.7 show that Land Grant Colleges are not correlated with the contemporaneous value of production, employment, capital invested, or the value of home production in manufacturing in 1850, and with additional measures in 1860, right before the passing of the Morrill Act.

19. Confirming this, as a placebo, in Online Appendix Table G.8 we run our core second-stage regression but using the 1890 instrument, revealing insignificant results throughout.

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unobserved correlation with engineers, as evinced by the LASSO results, presented next.

3.3. Robustness

We undertake several additional robustness exercises. First, Table 7, columns (1) and (2), uses LASSO regression (Belloni, Chernozhukov, and Hansen 2014) to impose additional discipline on variable selection, offering a more parsimonious representation of the most important explanatory variables. Column (1) presents the results for OLS and column (2) for IV estimations. The LASSO procedure sets virtually all geographical controls, slavery, secondary education, and physicians to zero. However, core structural variables—manufacturing output, railroad infrastructure, farms per capita—as well as basic literacy and higher-order human capital variables— share with tertiary education, and lawyers—remain critical. The magnitude of the engineers variable is barely changed.[20]

Second, there could be *other* reasons why Land Grant College towns were economically special besides generating engineers. To explore this possibility, we repeat our exercises, dropping the counties where the Land Grant Colleges are located, to mitigate the impact of the university and associated activities per se, leaving the results essentially unchanged (Online Appendix Table G.9).[21] As a further exercise and to address any possible selection bias of the Land Grant Colleges locations, we follow Andrews (2021) in restricting the sample only to those counties that were formally in consideration for receiving one.[22] This guarantees that the comparison group is more similar to the actual Land Grant Colleges receivers along the unobservables correlated with the selection of sites, although it substantially reduces the sample. Table 8 shows that OLS regressions with this reduced sample show very similar results to those above, while the instrumented regressions confirm the significant, positive impact of engineers.[23] The effect is not present for lawyers or doctors, consistent with the baseline findings in the full sample.

To relate the county-level results with the state results in the next section, we replicate the exercise for the 1900 5% census subsample at the state level, with the number of covariates reflecting the reduced degrees of freedom.[24] This higher level of

20. Online Appendix Table G.11 reports the estimates without state fixed effects, which are slightly larger.

21. Our county-level estimates are also consistent for subsamples of regions such as the South, New England and non Rust Belt areas (consistent with Franck and Galor (2021)).

22. We thank Michael Andrews for sharing with us the data of his ongoing project. Andrews (2021) shows the impact of colleges on local innovation and we can think of this exercise as tracing the intermediate impact through engineering/adoptive technical capacity and then on to income. This subsample of counties is richer, more educated and produced more innovations to start with, consistent with the findings in Andrews (2021).

23. Online Appendix Table G.10 shows the results for running the same regression, without controlling for patents.

24. We thank Ufuk Akcigit for suggesting this exercise.

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TABLE 7. Robustness: LASSO, non-engineers income and modern engineers (county level, United States).

	Dependent variable: Ln (income per capita in 2000)				
	(1) LASSO OLS	(2) LASSO IV	(3) OLS	(4) OLS	(5) OLS
Engineers	0.0881***	0.107***	0.104***	0.0943***	0.0823***
	(0.0219)	(0.0203)	(0.0308)	(0.0267)	(0.0278)
Rainfall			0.0446	0.0945	0.127
			(0.100)	(0.0863)	(0.0908)
Altitude			−0.455**	−0.207	−0.187
			(0.189)	(0.147)	(0.142)
Ruggedness			0.0546	−0.0891	−0.0108
			(0.0884)	(0.0813)	(0.0781)
River distance			0.0375	−0.238	0.509
			(0.261)	(0.226)	(0.334)
Temperature			−0.0422	0.0391	−0.0360
			(0.176)	(0.180)	(0.144)
Coast distance			−0.261	−0.218	−0.0687
			(0.165)	(0.160)	(0.118)
Population density			0.0239***	0.0152**	0.0957***
			(0.00489)	(0.00651)	(0.00656)
Manufacturing output	0.140***	0.145***	0.0629**	0.109***	0.0657*
	(0.0338)	(0.0344)	(0.0304)	(0.0277)	(0.0372)
Slavery			0.0985	0.0199	0.148**
			(0.0635)	(0.0522)	(0.0629)
Railroads	0.136***	0.118***	0.0863***	0.119***	0.0505***
	(0.0211)	(0.0214)	(0.0234)	(0.0251)	(0.0173)
Farms per capita	−0.0141	−0.0476*	0.00613	−0.0414	−0.0198
	(0.0256)	(0.0248)	(0.0326)	(0.0262)	(0.0371)
Literacy	0.390***	0.390***	0.299***	0.210***	0.426***
	(0.0482)	(0.0484)	(0.0630)	(0.0493)	(0.0722)
Secondary			0.0312	0.0406	0.0241
			(0.0497)	(0.0462)	(0.0482)
Tertiary	0.0867**	0.0921***	0.0145	0.0228	0.0157
	(0.0337)	(0.0336)	(0.0264)	(0.0242)	(0.0201)
Lawyers	0.118***	0.130***	0.0236	0.0259	0.211***
	(0.0265)	(0.0268)	(0.0537)	(0.0580)	(0.0517)
Physicians			−0.0251	0.0492	0.0221
			(0.0942)	(0.104)	(0.0944)
Modern engineers					0.391***
					(0.0367)
Observations	1,904	1,904	1,699	1,751	1,751
R-squared			0.130	0.159	0.390
State FE	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variables are: log sub-national income per capita (2000) (columns (1), (2), and (5)), log sub-national mean income for non engineers in 2010 (column (3)), and log sub-national mean income for managers in 2010 (column (4)). Engineering density is measured by engineers per 100,000 male workers. Patents density is measured by patents per 100 inhabitants. Rainfall captures total yearly rainfall in meters, altitude measures the elevation of the capital city of the state in kilometers, ruggedness of Terrain from Nunn and Puga (2012), and rivers capture the distance between the near river and the county. Rivers taken from HydroSHEDS (USGS 2011). Temperature is a yearly average in degrees celsius; distance to coast is the distance between the coast and the county centroid. Population density is the number of individuals per 100 square kilometers in 1880. Manufacturing output is the manufacturing output per capita in 1880, taken from NHGIS. Slavery to county level is taken from Nunn (2008). Farms per capita is the number of farms per capita retrieved from the University of Virginia Library. Railroads is an identifier variable if the county has a railroad. Literacy is the share of the population that was literate in 1880. Secondary is the share of 14–17-year-olds who report that they are attending school. Tertiary is the percentage of workers who report completing one or more years of college. Lawyer and physician density is measured per 100 individuals. Modern engineers is the proportion of employed in architecture and engineering in a county from 2010 census. Beta coefficients. Robust and clustered SE in parenthesis. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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TABLE 8. Andrews (2019) colleges and losing finalists (county level, United States).

Dependent variable: Ln (income per capita in 2000)					
	(1) OLS	(2) First stage	(3) First stage	(4) First stage	(5) IV
Engineers	0.137*				0.649*
	(0.0698)				(0.383)
Patents	0.00761	−0.00324	−0.00284	0.000438	0.0163
	(0.0159)	(0.0264)	(0.0146)	(0.00398)	(0.0146)
Rainfall	−0.138	−0.0679	−0.126	0.0403	−0.104
	(0.101)	(0.179)	(0.101)	(0.0489)	(0.146)
Altitude	−0.808**	0.423	0.651***	0.228**	−1.029**
	(0.339)	(0.381)	(0.236)	(0.0994)	(0.427)
Ruggedness	0.323*	0.0769	−0.0764	−0.0505	0.292
	(0.168)	(0.150)	(0.0974)	(0.0379)	(0.185)
River distance	0.757	−1.547	−1.394*	0.177	1.535
	(1.187)	(1.021)	(0.724)	(0.258)	(1.382)
Temperature	0.0963	0.709***	0.369***	0.101***	−0.255
	(0.137)	(0.141)	(0.0855)	(0.0280)	(0.294)
Coast distance	−0.0777	0.348**	0.244***	0.162***	−0.243
	(0.101)	(0.144)	(0.0873)	(0.0310)	(0.185)
Population density	−0.388***	−0.107	−0.0226	0.0381***	−0.321***
	(0.0440)	(0.0834)	(0.0417)	(0.0125)	(0.0683)
Manufacturing output	0.0672	0.134	−0.0413	−0.0132	−0.0192
	(0.0638)	(0.0931)	(0.0417)	(0.0152)	(0.0984)
Slavery	0.0542	0.0322	0.241**	0.0601	0.0531
	(0.211)	(0.167)	(0.120)	(0.0434)	(0.217)
Railroads	0.0652	0.109	0.0909*	0.0546***	−0.00278
	(0.107)	(0.0903)	(0.0543)	(0.0205)	(0.119)
Farms per capita	−0.397***	−0.603***	−0.435***	−0.0238	−0.0930
	(0.116)	(0.114)	(0.0728)	(0.0265)	(0.270)
Literacy	0.585**	0.500**	0.631***	0.236***	0.346
	(0.246)	(0.228)	(0.158)	(0.0594)	(0.321)
Secondary	−0.297*	0.127	−0.00367	0.00604	−0.348*
	(0.172)	(0.141)	(0.0931)	(0.0403)	(0.193)
LGC 1862		0.0742**	0.00593	0.00492	
		(0.0314)	(0.0146)	(0.00572)	
Observations	202	202	204	204	202
R-squared	0.420	0.469	0.458	0.377	0.251

Notes: Dependent variable is log sub-national income per capita (2000). Standardized beta coefficients. Data are restricted to the Andrews (2019) sample. Engineering density is measured by engineers per 100,000 male workers. Patents density is measured by patents per 100 inhabitants. Rainfall captures total yearly rainfall in meters, altitude measures the elevation of the capital city of the state in kilometers, ruggedness of Terrain from Nunn and Puga (2012), and rivers capture the distance between the near river and the county. Rivers taken from HydroSHEDS (USGS 2011). Temperature is a yearly average in degrees celsius; distance to coast is the distance between the coast and the county centroid. Population density is the number of individuals per 100 square kilometers in 1880. Manufacturing output is the manufacturing output per capita in 1880, taken from NHGIS. Slavery to county level is taken from Nunn (2008). Farms per capita is the number of farms per capita retrieved from the University of Virginia Library. Railroads is a identifier variable if the county has a railroad. Literacy share of the population that is literate in 1880. Secondary is the share of 14–17-year-olds who report that they are attending school. Engineers instrumented using a dummy for whether the county had a Land Grant college or not, estimated by 2SLS. Robust and clustered SE in parenthesis. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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aggregation also allows for more labor mobility among units of analysis, as discussed by Aghion et al. (2009). In theory, this could reduce the impact of our instrumented engineering variable but, again, it does not seem to prevent the emergence of a very robust and significant coefficient. It also allows introducing a measure of mining activity that is available at the state level. The OLS and IV estimates at 0.1 and 0.2 (Online Appendix D Tables G.12 and G.13) are significant and very close to the corresponding county-level estimates, respectively.

Taken together, the US data suggest that technical capacity, spanned by engineering and patenting density, has a strong and independent effect on future income levels. In our preferred specification, the beta coefficient for engineering is 11%. Though not instrumented, patenting capacity leads to another 11%. Hence, higher level of historical technical human capital plausibly account for substantial differences in modern income per capita. This does not imply the unimportance of lower level human capital: the beta coefficient (again uninstrumented) for literacy is 50%. A comparable increase in secondary schooling leads to only a 4.5% increase, although, again, the high school movement began substantially later and thus the variance in 1880 is very small. We analyze the possible mechanisms of transmission of the sizable technical capacity effect next, and we present estimates with international data in Section 5.

4. Mechanisms of Influence and Persistence

What are the mechanisms through which engineering or technical capabilities affect income today? We approach this question in several ways. First, Table 7 explores whether the effect is somehow limited to higher-order human capital as an input, or whether as a proxy for broader technical capabilities, it has spillovers beyond the impact on engineers per se. Column (3) offers a first approximation showing that engineering density has a similar impact on modern income when incomes of engineers are *excluded*, suggesting our results are not merely compositional. Column (4), for instance, shows a similar impact on the incomes of managers as a class, suggesting there is a broad-based rising of the profitability of firms in higher engineering density areas, as in Murphy, Shleifer, and Vishny (1991). Column (5) shows that the channel of transmission is *not* just through a trivial correlation of historical engineering density with modern engineering density, which drives today's income. The coefficient for engineers falls only slightly when controlling for their modern-day counterparts. Together, these results suggest that historical engineering density is in fact capturing technical capacity at the turn of the 20th century, which would have long-lasting impacts on the broader economy.

As an illustration of the time path of influence, Figure 2 plots the coefficient of engineering density on county-level manufacturing value added at 20 year intervals from 1860 to our midpoint of 1940, after which the censuses do not tabulate this information in a comparable fashion.[25] We see that the 1880 engineering density

25. For instance, GDP per capita statistics start in 1950.

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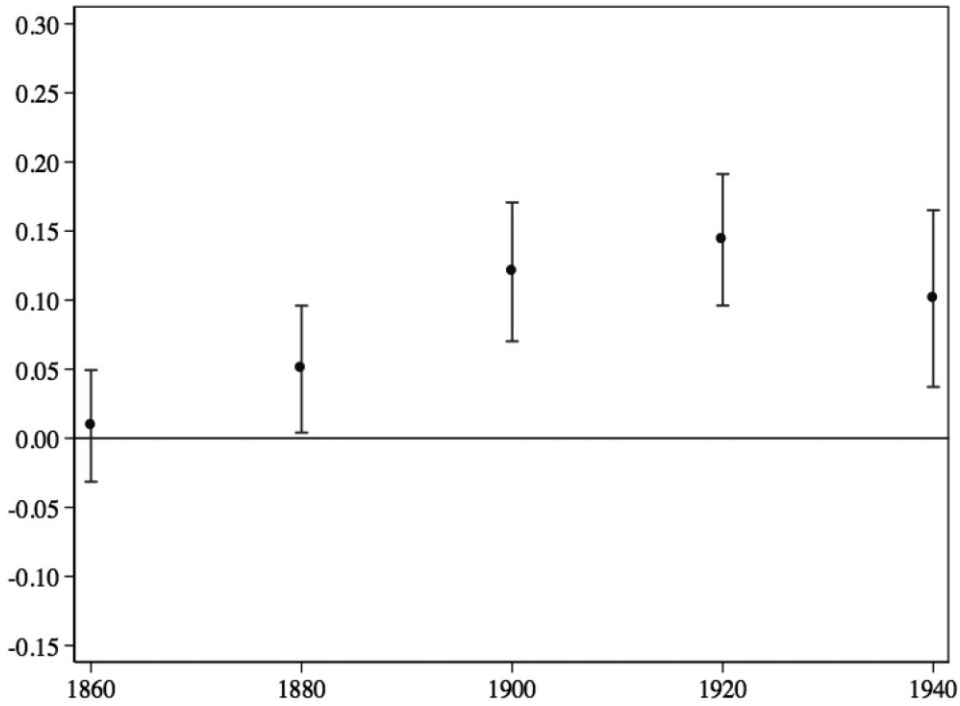


FIGURE 2. Engineering density on US manufacturing value 1860–1940. The figure plots the coefficient of engineering density on manufacturing value added at 20 year intervals from 1860 to 1940, at the county level, for the United States. Controls included are mentioned then. Rainfall captures total yearly rainfall in meters, altitude measures the elevation of the capital city of the state in kilometers, ruggedness of Terrain from Nunn and Puga (2012), and rivers captures the distance between the near river and the county. Rivers taken from HydroSHEDS (USGS 2011). Temperature is a yearly average in degrees celsius; distance to coast is distance between the coast and the county centroid. Population density is the number of individuals per 100 square kilometers in 1880. Slavery to county level is taken from Nunn (2008). Farms per capita is the number of farms per capita retrieved from the University of Virginia Library. Railroads is an identifier variable if the county has a railroad. Beta coefficients. Robust and clustered SE reported. Lines mark 95% confidence intervals.

measure had an insignificant impact in 1860, alleviating concerns about pre-trends. But from 1880 to 1920, its coefficient grew steadily in magnitude and significance. The final 1940 point, while decreasing slightly in value, is statistically of the same magnitude.[26] Hence, we have an enduring influence of 1880 engineering density on manufacturing growth well into the 20th century.

The adoption of existing technologies drives higher incomes across the long run through both increased productivity in existing sectors and the structural evolution of the economy into new sectors. The county-level findings that engineers had an impact

26. Data for 1930, used in Table 9, show the same pattern and we exclude it only for simplicity in the graph.

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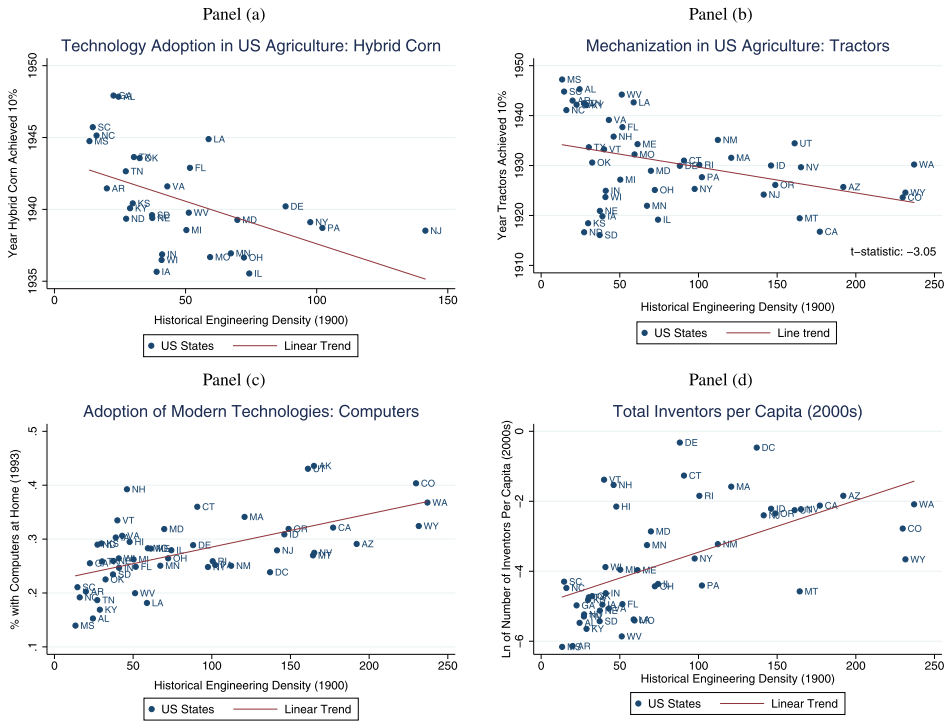


FIGURE 3. Engineering density and adoption of different technologies (US states). US state level engineering density in 1900 versus (a) year in which hybrid corn cropland reached 10%, (b) year in which tractor use achieved 10%, and (c) the percentage of the population living in homes with personal computer in 1993. All data are from Skinner and Staiger (2007).

beyond inventions captured in patenting suggest that a critical channel of influence is exactly facilitating this adoption process. At the state level, engineering density shows a significant positive correlation with several celebrated technology adoption measures (Griliches 1957; Skinner and Staiger 2007): the adoption rates of hybrid corn; rates of mechanization in agriculture proxied by the introduction of tractors; and, in the digital age, the percentage of homes with personal computers in 1993 (Figures 3(a)–(c)). Figure 3(d) shows it is also correlated with the density of inventors today, suggesting that the engineering vocation of a region supported higher-order scientific pursuits over the long run.

Table 9 presents seven indicators drawn from the 1930 manufacturing census, one of the earliest and most complete, undertaken 50 years after our engineering measure—roughly half way through our historical span. Three are measures of industrialization: manufacturing value added per capita and manufacturing establishments per capita capture structural transformation, while horsepower per firm is more likely to capture the increasing technological intensity of production. Following the association made by Murphy, Shleifer, and Vishny (1991) of engineering with entrepreneurship and

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TABLE 9. Intermediate mechanisms through which engineering affects GDP (1930, county level, United States).

	(1) *Value manufacturing per capita*	(2) *Horsepower in manufacturing*	(3) *Establishment per capita*	(4) *Retail stores per capita*	(5) *Wholesale sales per capita*	(6) *Bank deposits per capita*	(7) *Banks*
Engineers	0.0688**	0.0464**	0.0611**	0.0794**	0.0937***	1.559**	0.0145**
	(0.0258)	(0.0214)	(0.0271)	(0.0344)	(0.0298)	(0.687)	(0.00702)
Observations	1,745	1,722	1,745	1,884	1,761	1951	1951
R-squared	0.490	0.514	0.486	0.470	0.419	0,153	0,199
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is the log of: value of manufactured products (column (1)), horsepower in manufacturing (rated capacity of prime movers of electric motors) (column (2)), manufacturing establishments (column (3)), retail distribution stores (column (4)), and net sales of wholesale establishments in thousands of dollars (column (5)). Bank deposits per capita and bank branches in columns (6) and (7). Engineering density is measured by engineers per 100,000 male workers. Patents density is measured by patents per 100 inhabitants. Rainfall captures total yearly rainfall in meters, altitude measures the elevation of the capital city of the state in kilometers, ruggedness of Terrain from Nunn and Puga (2012), and rivers capture the distance between the near river and the county. Rivers taken from HydroSHEDS (USGS 2011). Temperature is a yearly average in degrees celsius; distance to Coast is the distance between the coast and the county centroid. Population density is the number of individuals per 100 square kilometers in 1880. Manufacturing output is the manufacturing output per capita in 1880, taken from NHGIS. Slavery to county level is taken from Nunn (2008). Railroads is an identifier variable if the county has a railroad. Farms per capita is the number of farms per capita retrieved from the University of Virginia Library. Literacy share of the population that is literate in 1880. Secondary is the share of 14–17-year-olds who report that they are attending school. Tertiary is the percentage of workers who report completing one or more years of college. Lawyer and physician density is measured per 100 individuals. $^{**}p < 0.05$, $^{***}p < 0.01$.

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industry-related services, we also employ as a proxy the number of retail stores per capita and value of net wholesale transactions per capita (see also Glaeser, Kerr, and Kerr 2015). Finally, we have two measures of financial depth, bank deposits and banks per capita, which may capture how technological upgrading gave impetus to a service industry to finance it (Jaremski 2014; Fulford 2015).[27] In all cases, engineers emerge as statistically significant with the full controls set, highlighting the importance of technical capacity for structural transformation and transition into more advanced sectors.

As a final check that the correlations with hybrid seeds and tractor adoption above are not pointing to the engineering variable picking up some more generic modernizing factor affecting all sectors—and analogous to the discussion of lawyers above—Online Appendix Table G.4 considers the interaction of engineers with agricultural activity (as proposed by Kantor and Whalley (2019)).[28] When we split the sample (using the median number of farms per capita in 1880), the effect of engineers in both OLS and IV estimates on long-term income is concentrated in *less* agricultural areas. Though technical change in agriculture was an important precursor for growth and engineering capabilities facilitated that, overall, our engineering measures appear to have worked more through structural transformation, developing manufacturing and perhaps associated services, as well as within-industry productivity.

Though the census will not permit carrying this exercise to the present day, recent specialized data sets confirm the long-run cumulative impact of engineering capacity. Using the County Business Patterns (2012) data set collected by the US Census Bureau, Table 10, columns (1–3), confirm the correlation of 1880 engineering density with the number of high-tech industries in a county, as well as payroll and employment in these sectors. This measure is also correlated with the incidence of Knowledge Intensive Business Services (KIBSs) employed by Cermeño (2019) in 2010 and 1980 (columns (5) and (6)), but also historically—as early as 1930 (in column (4)). Across this 80 year span, the coefficient on 1880 engineering density remains remarkably constant, speaking to the persistence of the human capital effect. Finally, using the Lexis/Nexis Ci Technology Database (2016), which provides technology marketers an insight into the installed technology and purchase plans for 120,000 US businesses, we are able to document the importance of technology investments at the intensive margin within firms. Columns (7–9) show that 1880 engineering density is significantly correlated with the number of personal computers per firm, the number of servers, and the budget allocated by factories for information technology, 130 years later.

Section 5 revisits the relationship between historical engineering density and present income using international data, both at the sub-national and national levels. Section 6 then draws on historical case studies as another means to document the channels through which engineering density affects growth.

27. Results also hold when using 1920s finance data instead.

28. We thank a referee for suggesting this sample split.

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TABLE 10. Modern mechanisms through which engineering affects GDP (county level, United States).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	High-tech sectors (2012)			Knowledge intensive business sectors			Investments in technology (2016)		
	Payroll	Employment	Establishments	1930	1980	2010	PCs	Servers	IT budget
Engineers	82.87*	0.917**	0.995**	0.000762***	0.00137***	0.000800**	0.0525***	0.00857**	22.40***
	(44.83)	(0.43)	(0.37)	(0.0002)	(0.0003)	(0.0003)	(0.017)	(0.003)	(7.11)
Observations	1.754	1.754	1.754	1.854	1.799	1.853	1954	1954	1954
R-squared	0.132	0.158	0.192	0.162	0.189	0.195	0.106	0.09	0.084
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Independent variable is: Engineering density is measured by engineers per 100,000 male workers. Data on high-tech sectors are from the County Business Patterns (2012). We use data on payroll, employment, and the number of establishments in these sectors (columns (1)–(3)). Knowledge Intensive Business Sectors (KIBS) for 1930, 1980 and 2010 in columns (4–6) are taken from Cermeño (2019). Investments in technology are taken from the Ci Technology Database (2016). We use the number of personal computers, servers, and IT budget per capita (columns (7)–(9)). $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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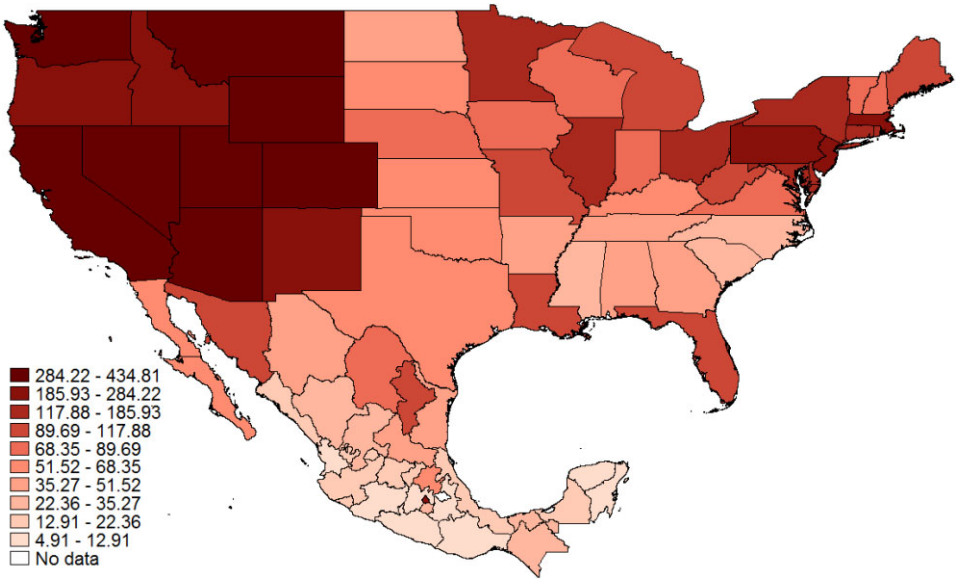


FIGURE 4. Sub-national engineering density, United States and Mexico, in 1900. Engineering density at the sub-national level for North America derived from census reported engineers per 100,000 male workers around 1900.

5. International Data

5.1. International Sub-National Engineering Data

To test the external validity of our findings, we undertake similar analyses at the sub-national (state) level for Argentina, Chile, Colombia, Mexico, Venezuela, and the United States, for which historical census data are available at this level of disaggregation. To give a feel for the disparities, Figure 4 maps this data for the United States and Mexico by decile of engineering density. Perhaps predictably, the advanced New England states and the heavily mining dependent and generally less populated Western states show the highest density, while the emerging industrial centers of the Midwest are close behind. The South is concentrated in the lower ranks, with South Carolina, Georgia, Arkansas, and Alabama in the bottom deciles—the lowest density in the United States. What is striking, however, is that the country that was the principal mining center of the Spanish empire (Mexico) is almost entirely concentrated in the first and second quintiles, with Sonora and the two Baja Californias appearing in the third and fourth deciles. Taking out Mexico City and the border states, Mexico is almost uniformly below even the US South in density of engineers. As we discuss next, despite four centuries of mining, this country had not acquired a corpus of trained professionals in the field compared to relative newcomer, the American West. The other Latin American countries show similar patterns.

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TABLE 11. Technical capacity as a determinant of income (state level, Americas).

Dependent variable: Ln (income per capita in 2000)						
	(1)	(2)	(3)	(4)	(5)	(6)
Engineering	0.1***	0.1***	0.07***	0.1***	0.1***	0.05**
	(0.04)	(0.03)	(0.02)	(0.03)	(0.02)	(0.02)
Population density		0.07**	0.06***		0.1***	0.06***
		(0.03)	(0.02)		(0.04)	(0.01)
Literacy			0.4***			0.4***
			(0.13)			(0.10)
Area				−0.03	0.08	−0.04
				(0.06)	(0.05)	(0.04)
Ruggedness				−0.03	−0.06	−0.04*
				(0.05)	(0.04)	(0.02)
Rainfall				−0.1*	−0.1	−0.09
				(0.06)	(0.08)	(0.06)
Altitude				−0.03	−0.07	−0.03
				(0.05)	(0.06)	(0.05)
Landlocked				−0.02	−0.003	−0.02
				(0.03)	(0.02)	(0.04)
Constant	0.6	0.6	0.6**	0.5	0.5	0.6**
	(0.42)	(0.40)	(0.28)	(0.39)	(0.37)	(0.23)
N	170	166	166	160	156	156
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.12	0.19	0.38	0.22	0.32	0.51

Notes: Dependent variable is log sub-national income per capita (2000). Data for Argentina, Chile, Colombia, Mexico, United States, and Venezuela. Engineering density is measured by engineers per 100,000 male workers. Population density is log number of individuals per square kilometer in 1900. Literacy is share of the population that is literate in 1900. Beta coefficients. Bootstrapped clustered SE in parenthesis. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 11 presents the results from the sub-national panel and comes to similar conclusions to those previous. Namely, we estimate

$$Y_{2005,ij} = \alpha + \gamma_E Eng_{1900_{ij}} + \gamma_{pop} Pop_{1900,ij} + \gamma_L Lit_{1900,ij} + \boldsymbol{Geo_i \gamma_{Ge}} + \mu_j + \varepsilon_{ij}. \quad (2)$$

We now take country, instead of state fixed effects. Engineering density enters persistently, significantly, and very close to the magnitude of the OLS estimates from the US county and state level samples in columns (1) and (2). In column (3), literacy again enters positively and significantly and reduces the coefficient on engineering by 30%, again broadly consistent with the US results in both effect and overall magnitude of the final engineering coefficient. Though we cannot control for a more complete set of human capital measures, the county-level exercises shown before suggest that, in fact, engineering is capturing innovation-related human capital and not human capital accumulation more generally. In sum, the estimates suggest that our county-level US

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findings resonate internationally: higher-order human capital related to engineering, science, and technology is an important determinant of long-run income.

5.2. Cross-Country Data

To explore the relationship at the highest level of aggregation, we return to Figure 1, which plots our measure of national engineering density against GDP per capita, both in 1900.[29] Though our primary focus is on the divergence within the Western Hemisphere, we bring in select European countries—the two colonial mother countries of Latin America, and two countries often paired with Latin America in income and natural resource intensive economic structure.[30] We first note that there is substantial variance in the stock of engineers. The Northern United States, with a density of 160, is the highest in our sample, roughly double the average for the country as a whole, 84, but still below the estimates by Ahlström (1982) for France (200) and Germany (250), the acknowledged leaders in science and engineering of the time (Online Appendix Figure F.1).

The US South shows about a third of the engineering density of the North at 60, consistent with the historical narrative (Wright 1986). However, lagging as it was, the US South was three times more advanced than the Latin American countries that average under 20 magnitudes, again, plausibly clustered slightly below those of the colonial mother countries. What is most striking is that countries that we tend to associate with a declining relative position across the previous century, especially Argentina and, to a lesser degree, Chile and Mexico, show densities a fraction of the two reference countries of very similar levels of income and structure, Denmark and Sweden, which would converge to the frontier.[31] The factor of five difference between the two peripheral areas—Scandinavia versus the Iberian peninsula—is consistent with the characterization of Landes (1998) and Bénabou, Ticchi, and Vindigni (2021) of their respective attitudes towards science and the Enlightenment. Both Sweden and Denmark's institutions of higher technical learning date from the 1700s, and Sweden's high density is consistent with the characterization by Sandberg (1979) of the country as the "Impoverished Sophisticate" that was exporting engineers to the United States at this time and contributing with consequential inventions.[32] This was not the case for Spain and Portugal.

29. Online Appendix Figure F.4 presents the relationship between modern engineers and income.

30. See Blomström and Meller (1991) and Ducoing and Peres-Cajías (2021).

31. Online Appendix Figure F.5 illustrates this point for two industrial manufacturing centers in our sample: Antioquia (Colombia's industrial hub) and New York state to which it would catch up only a century later.

32. The overproduction of engineers led many to emigrate to the United States and, in the 19th century, Swedish engineers are credited with inventing the blowtorch, ball bearings, ship propellers, the safety match, the revolver, the machine gun, dynamite, and contributing to the development of bicycles, steam turbines, early calculators, and telephony (Ericsson), among others. We look at the role of foreigners in Online Appendix Table G.17 and the control for foreign immigration in Online Appendix Table G.18.

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The data for their ex-colonies in Latin America are consistent with historical accounts. Safford (1976), in his classic *Ideal of the Practical*, observes that "Latin American societies in general, and the upper classes in particular, have been considered weak in those pursuits that North Americans consider practical, such as the assimilation, creation, and manipulation of technology and business enterprise in general" (p. 3). Graham (1981) argues that Brazil lagged far behind the American South in every aspect of industrialization, transportation, and agricultural technology. While agricultural journals abounded in the South, lower levels of literacy prevented this in Brazil. A long literature has focused on Argentina's weakness in innovation efforts in comparison to countries such as Australia and Canada, seen as similarly endowed (see, for example Diaz Alejandro 1985; Duncan and Fogarty 1986; Campante and Glaeser 2018). Canada's estimate below the US South probably understates its technical capacity since it was part of the greater New England scientific community. The four principal Canadian universities emitting graduates—McGill, University of Toronto, *Ecole Polytechnique* in Montreal, and Queen's University in Kingston, Ontario—lay within a circle of 350 mile radius with Cornell University at its center, and that includes many of the principal US departments of the time.[33] However, understated, it still has more than double the engineering density of Argentina, which is also supported by historical accounts. The determined efforts of both Canada and Australia to achieve widespread literacy in the prairies had no analogue in Latin America, nor did the extensive expansion in the form of experiment stations and technical assistance. As Online Appendix A documents, the national scientific establishments and professional training of civil engineers appeared much later and on a smaller scale. As an example, wealthy Argentina began graduating engineers only in 1870, and Peru, one of the premier mining centers in the hemisphere, in 1880, roughly on the same time line as Alabama. General engineering associations were set up in many countries around the same time that, in the United States, associations in individual sub-fields were established. Arguably, in Latin America, natural resource rents, while elevating income, were not being deployed as they were in the United States or Scandinavia for the development of technical capabilities that would prepare them for the next phase of industrialization. In the framework of Howitt and Mayer-Foulkes (2005), we have countries with similar levels of Schumpeterian backwardness, but with radically different levels of absorptive capacity.

Supporting this view (and showing the potential usefulness of our data), Alfaro and Comin (2019) show that the national estimates of engineering densities are negatively and significantly correlated with their measures of technological adoption lags, confirming the findings at the county and state levels in the previous sections (see Online Appendix Table G.14). Further, Online Appendix C and Online Appendix Table G.15 show that these differences in engineering densities for the countries in Figure 1

33. Although the first engineering classes graduated in the 1870s, engineering courses were in place by the 1850s. Further, the articulation of the different fields of engineering occurred only somewhat later than in the United States. The development of, for instance, mechanical engineering as a separate course occurred about 20 years after in the United States, and electrical engineering 10–15 years after.

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are indeed significantly and positively correlated with present income. As a back of the envelope calculation, applying a coefficient of 0.1 as a rough consensus value from the United States and international sub-national exercises can account for roughly a quarter of the cross-country variation in incomes.

6. Support from History: Case Studies

To flesh out our understanding of the mechanisms further, the next section presents several case studies from the US South and Latin America that document stunted long-run within-sector productivity growth and frustrated structural transformation as a result of deficient technical capacity.

6.1. The US South

The United States started relatively early and energetically in the training of engineers, but almost entirely in the North and Midwest. The first institution of engineering education emerged from the Revolutionary War at West Point, established in 1802, which trained engineers for both military and civilian purposes. By 1862, there were roughly a dozen engineering schools in the East, but there were also schools as far west as Michigan and as far south as Maryland.[34] The passage of the Morrill Land Grant Act in 1862, used in Section 3.2, led to an acceleration in the establishment of engineering programs, roughly sextupling the number in the decade after passage. The Act led to the establishment of the Columbia School of Mines in 1864, Worcester Polytechnic in 1868, the Thayer School of Civil Engineering at Dartmouth College in 1867, and Cornell University, as well as new universities in Iowa, Nebraska, Ohio, and Indiana. Also a land-grant college, MIT, was founded in 1861 in response to the increasing industrialization of the United States, adopting a European (German) polytechnic university model stressing laboratory instruction in applied science and engineering. The period also saw a deepening, as the profession in the US diversified further into sub-branches. The Polytechnic College of the State of Pennsylvania, for instance, founded in 1853 and inspired by the polytechnic schools in Europe, granted degrees in Mechanical Engineering in 1854 and Mining Engineering in 1857. The establishment of professional societies in civil (1852), mining (1871), mechanical (1880), and electrical (1884) engineering testifies to the consolidation of a process of specialization and diversification. By 1890, a modern and world-class engineering profession was firmly established.

The South had a different trajectory. Engineering programs were few before the Civil War and secession both removed southern opposition to the passing of the

34. The American Literary, Scientific, and Military Academy at Norwich, Vermont awarded its first Civil Engineering degrees in 1837, and Rensselaer School in New York, in 1835. Harvard established the Lawrence Scientific School in 1846, Brown offered its first engineering course in 1847, Dartmouth began the Chandler Scientific School in 1852, and Yale began the Sheffield Scientific School in 1846.

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Morrill Act and ensured that the southern states would not benefit from the first wave of colleges.[35] The training of engineers began seriously only after the Civil War, with special impetus from the second phase of the Morrill Program in 1890. The storied Georgia School of Technology (now Georgia Tech) was established in 1888, patterned on MIT, Worcester Polytechnic Institute (WPI), and other technical schools, and instigated by two confederate generals who acutely felt the need to be technologically competitive with the North and transform southern society.

However, this half-century late start had lasting implications for technology adoption and convergence of the region with the North. The emblematic failure was the Birmingham Steel industry, referred to in the introduction. Central was that the low iron, high phosphorous nature of Alabama red hematite required substantial adaptations of technology to the southern context, which the local technical capacity was not able to engineer. Nor was it able to develop southern versions of new inventions in the paper and textile industries. In lumbering and iron making, as well, southern producers of the 1920s were not only *not* innovative, they were using methods phased out decades earlier elsewhere. Wright (1986) casts much of his work explaining the US South's persistent lag exactly in terms of an adoptive capacity framework. "The fundamental reason for [the protracted lack of uptake of technologies] is that early industrialization is a matter of learning in the broadest sense of that term... Having missed the formative phases of the 'American System', the South was lacking a machine-tools and capital-goods sector almost entirely and therefore was bypassed by the kind of adaptive, dynamic, path-breaking series of technological breakthroughs that made the 'American System' distinctive" (pp. 124–125). Arguably, the big push by the federal government, ranging from the Land Grant colleges to the selective location of advanced industries that encouraged the migration of higher-order human capital, raised technical capacity toward the frontier, and permitted catch up over the course of the 20th century.

6.2. Latin America versus the United States

The catastrophic impact of Latin America's late start is nowhere more in evidence than in the industry in which the region for centuries had had a true comparative advantage, yet by the turn of the 20th century had lost—mining—which the Second Industrial Revolution radically transformed. In perhaps the starkest case, Chile saw its world market share of copper fall from 40% to under 4% by 1911 (Online Appendix Figure F.6), and even as early as 1884, the *Sociedad de Minería* (Mining Association) wondered openly whether Chile's copper mines would survive at all (Collier and Sater 1996). Chilean historians date this technological slippage to the beginning of the 19th century, when "the work of mining was not very systematic" and the "receipt

35. Auburn University in Alabama began its engineering program in 1872, and North Carolina State in 1887. Texas A&M awarded its first degree in Civil Engineering in 1880, Virginia Tech in 1885 in Mining Engineering, and the University of Kentucky, although having an engineering program dating from 1869, graduated its first civil engineer in 1890. As early as 1838, the University of Tennessee was teaching courses in Civil Engineering, and in 1879 it began awarding doctorate degrees in Civil and Mining Engineering.

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of industrial innovations [from abroad] was slow and without visible influence" (Villalobos and Beltrán 1990, pp. 95–96). Charles Lambert, a representative of a British mining company in La Serena who was trained in the *École Polytechnique in Paris*, noted in 1819 the primitive mining practice, scarce knowledge of minerals, and inefficient smelting, all of which represented poor technique relative to that employed in Europe (Maloney 2007).[36] Only with the purchase of the mines by United Kingdom and United States firms did Chilean mining modernize, reaching new production highs. By 1918, American interests controlled 87% of Chilean copper output (O'Brien 1989).

Similarly, in Mexico, local entrepreneurs lost share in the mining industry they had dominated for centuries precisely due to their lack of capacity to master emerging technologies (Ruiz Larraguivel 2004; Brading 1971; Marichal 1997). As Figure 4 showed, in Mexico, even when including Zacatecas, San Luis de Potosí, and Guanajuato—long centers of mining—density was at low levels compared to the newcomers in the US West. Around 1900, abandoned, under-exploited, and newly discovered mines fell into foreign hands that could bring new global technologies to bear. The US grew to control close to 80% of investments in the sector (Maloney and Zambrano 2021). This phenomenon was visible throughout the continent. In Upper Peru (now Bolivia), the decline of silver production in some of the most famous mines, like those at Potosí, arose from the "failure to apply new mining techniques, heavy mortality among Indian laborers and the exhausting of previous rich veins" (Scobie 1964, p. 59). In Ecuador, Hurtado (2007) argues that the discovery of new mineral deposits was hindered by a resistance to scientific methods.

Given the similarly low engineering densities, it is perhaps not surprising that Spain's own development path shows the exact same issues of inability to manage new technologies and external dominance characterizing Latin America (Tortella Casares 2000). For example, the evolution of the Spanish mercury mining industry precisely foreshadows the subsequent experience in Latin America discussed earlier. Though Spanish mines were rich and mercury had been worked for a millennium, the lack of technical capacity, capital, and the slow growth of domestic metallurgical know-how led Spanish entrepreneurs to work close to the surface and sell out to foreigners once easy veins had been exhausted. In 1873, an Anglo-German led conglomerate purchased the copper mines on the Rio Tinto river in Andalucia, introduced new technologies, and from 1877–1891, became the world's largest producer of copper, contributing the to fall in Chile's global share in Online Appendix Figure F.6. As Tortella Casares (2000, p. 96, 213–215) summarizes: "extraction and processing constitute a classic example of the failure of Spanish entrepreneurs to confront the problems of developing

36. One of Chile's most venerated historians, Francisco Encina, noted that "from the point of view of capital and of technical and administrative aptitude, the copper industry is as demanding as the most complicated manufacturing industry" (Encina 1972, p. 62). However, his studies revealed "an extraordinary economic ineptitude in the national population consequence of an education completely inadequate to meet the demands of contemporary life" (Idem, p. 17).

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an industrial sector with complex technology, intensive use of capital, [and] a fast-expanding horizon."[37]

The eventual American dominance of the Chilean copper and Mexican mining industries strikingly illustrates the road that could have been taken with the same homogeneous product. Not only does Wright (1999) argue that the United States in the 19th century "parlayed its [natural] resource-based industrial prosperity into a well-educated labor force, an increasingly sophisticated science-based technology, and world leadership in scientific research itself" (Wright 1987, p. 665), but he uses precisely the US copper industry as an example of national learning and of innovation as a network phenomenon. In the post-Civil War period, the United States became the foremost location for education in mining engineering and metallurgy. It was the revolution in metallurgy (e.g. the Bessemer process and the introduction of electrolysis on a commercial scale for the refining of copper) that propelled the copper industry during the last decades of the 19th century and laid the groundwork for much subsequent industrialization. The transference of these technologies by US firms to their mines and smelters in Chile and Mexico revolutionized the antiquated industries in both countries, dramatically increased production, and left them dominant in both cases. In Chile, Guggenheim's El Teniente mine was the first in the world to apply the flotation process in concentrating low-grade ores. Mechanizing digging made Chuquicamata in the north the largest open pit mine, and, again, a new concentration process was introduced using sulfuric acid and electrolytic precipitation to treat the mine's ore. From 1912–1926, copper production in Chile quintupled as a result, reversing a 25 year period of stagnation (O'Brien 1989). In Mexico, the US Guggenheim interests opened smelters in Monterrey (1892) and Aguascalientes (1894), purchased the largest Mexican Smelting and Refining company in 1906, introduced modern methods of extracting and refining silver ores, and, in addition, started the production of lead and zinc mining (Bernstein 1964; O'Brien 1989).

In sum, the historical evidence confirms that the radically different development outcomes attained in sectors ranging from steel to mining were importantly driven by the relative capabilities to adapt new technologies to the local context, capabilities which our engineering data show were weak in the American South, and virtually absent in Latin America.

7. Conclusion

Much of the growth literature stresses differences in the capacity to generate, import, and apply new technologies as central to explaining relative growth performance. To date, however, there had been little data generated on the higher level (technical)

37. Though beyond the scope of this work, we conduct some exploratory regressions on the drivers of innovation in Online Appendix E Tables G.16 and G.17. Consistent with the modern literature, religion emerges negatively and foreigners positively (Bénabou, Ticchi, and Vindigni 2015; Nunn, Sequeira, and Qian 2020).

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human capital, in particular, the central figure of the engineer, and none to attempt to verify its impact on income differentials. This paper collects data at various levels of aggregation to remedy that and show its importance for long-term growth.

It begins employing county-level data from the United States and establishes the robust explanatory power of engineering density and patenting on income, controlling for a variety of economic and geographic variables, measures of secondary and tertiary education, and attempting to control directly for potential endogeneity using the 1862 US Morrill Land Grant Colleges. The differences in the stock of engineers at the beginning of the Second Industrial Revolution substantially contributed to county income in the decades to follow. We then explore the impact of engineering density through several intermediate mechanisms across the next century at various intervals. Looking 50 years later—using the 1930 manufacturing census—we find a robust impact on manufacturing expansion, technological intensification, and structural transformation, and show that the impact on manufacturing value added increased steadily from 1860 to 1940. Engineering density is also correlated with the adoption of celebrated technologies such as hybrid corn and mechanization in agriculture. Its effects continue to the present day with high-tech sectors and technological intensity within sectors: with computer adoption, number of servers, and IT budget being positively correlated.

The paper then uses sub-national level and national-level data to explore the relationship internationally. It repeats the US exercises with a multi-country panel working at the state level, and finds similar orders of magnitude, confirming the importance of the upper tails of technical knowledge for long-term growth. Finally, it generates national-level engineering data for the Americas in 1900, which suggest that differences in technical capacity can plausibly explain around a quarter of the disparate development trajectories of countries which—at the time—had very similar levels of income. In the international data, the similarity of engineering density in Spain and Portugal to that of Latin America and the commonality of experiences with, for example, mining, suggest important inherited determinants of low capacity for technological absorption. Given the large impact we establish of higher-order technical human capital and inventive capacity on income, further investigation of their determinants offers a promising avenue of research for the growth agenda.

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Supplementary data

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